

# **Biofilm, Virulence Factors, CSH Analysis, Sequence Alignment, and Phylogenetic Tree of *Candida* Isolates**

By :Hasna Abbass

30/04/2026

## **Table of Contents**

- 1. Introduction
  - 1.1 Background
  - 1.2 Purpose of the Study
- 2. Objectives of the Project
- 3. Materials and Methods
  - 3.1 Dataset Description
  - 3.2 FASTA Sequences
  - 3.3 Tools and Libraries
  - 3.4 Data Processing Workflow
- 4. Python Code Used
- 5. Results
  - 5.1 Data Loading and First Output
  - 5.2 Biofilm Classification
  - 5.3 Virulence Score
  - 5.4 Biofilm/CSH Ratio
  - 5.5 Mean Values by Species
  - 5.6 FASTA Sequences and Alignment
  - 5.7 Distance Matrix and Phylogenetic Tree
  - 5.8 Graphical Results
- 6. Interpretation of Results
- 7. Conclusion

---

## **1. Introduction**

### **1.1. Background**

*Candida* species are opportunistic fungal pathogens that can cause a wide range of infections, particularly in immunocompromised individuals. Their ability to form biofilms, express

virulence factors, and adapt to different environments plays a critical role in their pathogenicity. In addition, molecular analysis such as DNA sequence comparison and phylogenetic studies provides insights into their genetic relationships and evolutionary patterns.

### 1.2. Purpose of the Study

This project aimed to analyze phenotypic and genetic characteristics of different *Candida* isolates using Python. The analysis included biofilm formation, metabolic activity, cell surface hydrophobicity, virulence genes, resistance mutation data, sequence alignment, genetic distance calculation, and phylogenetic tree construction.

The project used two types of data: a CSV file containing biological measurements for different *Candida* isolates and a FASTA file containing DNA sequences for selected *Candida* species.

---

## 2. Objectives of the Project

The main objectives were:

- To classify *Candida* isolates according to their biofilm production.
  - To calculate a virulence score based on the presence of the genes **ALS3** and **HWP1**.
  - To study the relationship between biofilm formation, metabolic activity, and cell surface hydrophobicity.
  - To compare mean values between *Candida* species.
  - To align DNA sequences and calculate genetic distances.
  - To construct a phylogenetic tree showing the relationship between *Candida* species.
  - To save the final calculated results in a new CSV file.
- 

## 3. Materials and Methods

### 3.1. Dataset Description

The dataset used in this study consists of a CSV file containing phenotypic and molecular data for different *Candida* isolates. The variables included:

- Biofilm\_OD570: measurement of biofilm biomass
- Metabolic\_OD490: metabolic activity
- CSH\_percent: cell surface hydrophobicity
- ALS3 and HWP1: virulence genes
- ERG11\_mutation: resistance-related mutation

These variables allow the evaluation of virulence and biofilm-forming capacity of *Candida* species.

### **3.2.FASTA Sequences**

The FASTA file (projectsequences.fasta) contains DNA sequences representing different *Candida* species. These sequences were used for alignment, distance matrix calculation, and phylogenetic tree construction.

The sequences vary in length and nucleotide composition, allowing comparison of GC content and genetic similarity between species.

### **3.3.Tools and Libraries**

The analysis was performed using Python. The main libraries used were:

- **pandas**
- **numpy**
- **matplotlib**
- **Biopython**

### **3.4.Data Processing Workflow**

The analysis followed these steps:

1. Data loading and cleaning
2. Conversion of variables to numerical values
3. Calculation of biofilm classification and virulence score
4. Statistical analysis and grouping by species
5. Sequence alignment and distance matrix calculation
6. Phylogenetic tree construction
7. Data visualization using graphs

---

## **4. Python Code Used**

(The complete code is presented and also integrated step-by-step in the Results section.)

---

## 5. Results

The results section presents the outputs obtained from the computational analysis, including numerical calculations, sequence analysis, and graphical representations.

### 5.1 Data Loading and First Output

#### Code used:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from Bio import SeqIO
from Bio.Align import MultipleSeqAlignment
from Bio.Phylo.TreeConstruction import DistanceCalculator,
DistanceTreeConstructor
from Bio import Phylo

df = pd.read_csv("data.csv", header=None)
# Split by comma
df = df[0].str.split(",", expand=True)

# Set first row as column names
df.columns = df.iloc[0]
df = df[1:].reset_index(drop=True)

# Clean spaces
df.columns = df.columns.str.strip()

print(df.head())
```

#### Output:

(table)

Example isolates:

| Species            | Biofilm OD570 | Metabolic OD490 | CSH (%) | ALS3 | HWP1 | ERG11 |
|--------------------|---------------|-----------------|---------|------|------|-------|
| Candida_albicans_1 | 1.15          | 0.92            | 72      | 1    | 1    | 0     |
| Candida_albicans_2 | 1.22          | 0.98            | 75      | 1    | 1    | 0     |
| Candida_albicans_3 | 1.05          | 0.88            | 70      | 1    | 1    | 0     |

#### Interpretation:

The dataset contains several variables for each isolate, including biofilm biomass measured at OD570, metabolic activity measured at OD490, cell surface hydrophobicity percentage, and the presence or absence of virulence or resistance markers. The first displayed rows correspond

to *Candida albicans* isolates, which show relatively high biofilm OD values and positive results for both ALS3 and HWP1 genes.

---

## 5.2 Biofilm Classification

### Code used:

```
df["biofilm_level"] = np.where(
    df["biofilm_OD570"] > 1,
    "Strong",
    "Weak"
)

print(df[["species", "biofilm_OD570", "biofilm_level"]])
```

### Interpretation:

This classification allowed the isolates to be separated into two biofilm levels. Isolates with higher OD570 values were considered stronger biofilm producers because OD570 reflects the biomass attached to the plate after staining.

---

## 5.3 Virulence Score

### Code used:

```
df["virulence_score"] = df[["ALS3", "HWP1"]].sum(axis=1)

print(df[["species", "ALS3", "HWP1", "virulence_score"]])
```

### Interpretation:

The virulence score provides a simple numerical estimation of the presence of selected virulence-associated genes. *Candida albicans* and *Candida tropicalis* showed the highest mean virulence score, indicating the presence of both ALS3 and HWP1 genes in these isolates.

---

## 5.4 Biofilm/CSH Ratio

### Code used:

```
df["biofilm_CSH_ratio"] = df["biofilm_OD570"] / df["CSH_percent"]

print(df[["species", "biofilm_OD570", "CSH_percent", "biofilm_CSH_ratio"]])
```

### Interpretation:

This ratio was used to compare biofilm formation with cell surface hydrophobicity. A higher value may indicate stronger biofilm formation relative to the hydrophobicity level. This calculation helped evaluate whether isolates with higher CSH also tended to form stronger biofilms.

---

## 5.5 Mean Values by Species

### Code used:

```
df["species_group"] = df["species"].str.replace(r"_\d+$", "", regex=True)

mean_by_species = df.groupby("species_group") [
    ["biofilm_OD570", "metabolic_OD490", "CSH_percent", "virulence_score"]
].mean()

print(mean_by_species)
```

### Interpretation:

The mean values showed clear differences between species. *Candida albicans* had the highest mean biofilm formation, followed by *Candida tropicalis*. These two species also showed the highest mean virulence score. In contrast, *Candida glabrata* showed the lowest mean biofilm formation and the lowest CSH percentage.

---

## 5.6 FASTA Sequences and Alignment

### Code used:

```
records = list(SeqIO.parse("projectsequences.fasta", "fasta"))

for record in records:
    print(record.id, record.seq)

alignment = MultipleSeqAlignment(records)
print(alignment)
```

### Interpretation:

The alignment allowed comparison of DNA sequences between species. This step was necessary before calculating genetic distances and constructing the phylogenetic tree. The

sequences were aligned in order to identify similarities and differences between the species at the nucleotide level.

---

## 5.7 Distance Matrix and Phylogenetic Tree

### Code used:

```
calculator = DistanceCalculator("identity")
distance_matrix = calculator.get_distance(alignment)
print(distance_matrix)

constructor = DistanceTreeConstructor()
tree = constructor.nj(distance_matrix)

Phylo.draw_ascii(tree)
Phylo.draw(tree)
```

### Interpretation:

A smaller distance value indicates that two sequences are more similar. For example, *Candida albicans* and *Candida tropicalis* had a small genetic distance, suggesting that they were relatively close in this sequence comparison. *Candida auris* and *Candida krusei* also showed a small distance, suggesting closer similarity between them in the analyzed sequence.

---

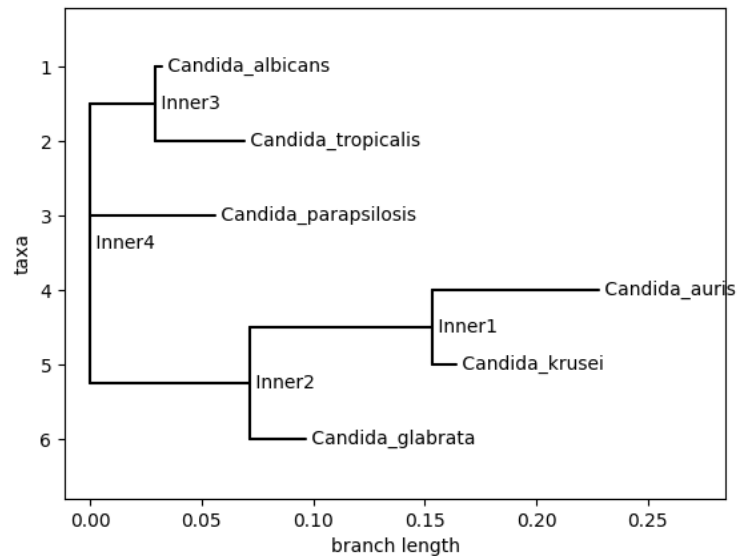
## 5.8 Graphical Results

---

### Figure 1. Phylogenetic tree of selected *Candida* species

### Code used:

```
Phylo.draw(tree)
```



### Interpretation:

The phylogenetic tree showed that *Candida albicans* and *Candida tropicalis* clustered close to each other, suggesting sequence similarity. *Candida auris* and *Candida krusei* also appeared close together. *Candida glabrata* was placed near the *C. auris*/*C. krusei* branch but separated from them, while *Candida parapsilosis* was positioned separately from the other groups.

---

## Figure 2. Biofilm biomass vs metabolic activity

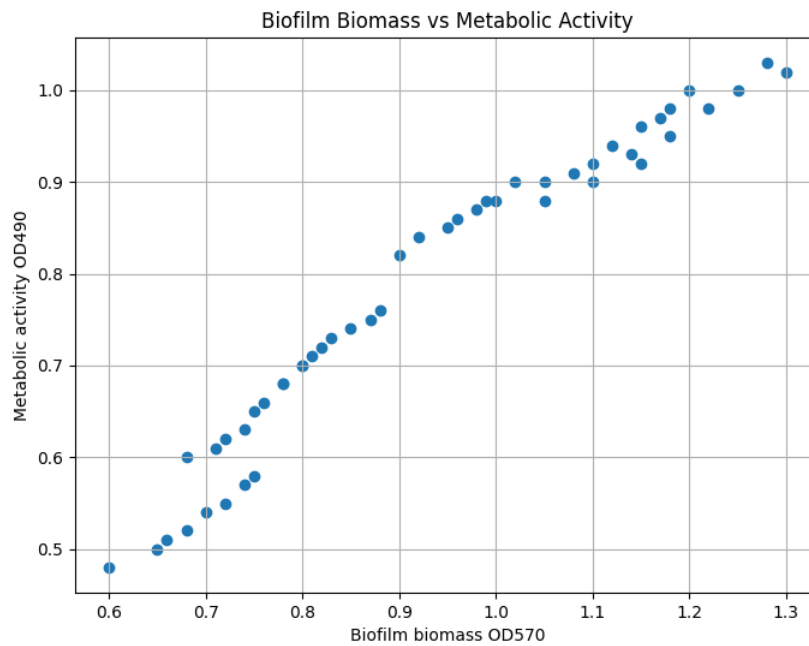
### Code used:

```

plt.figure(figsize=(8,6))
plt.scatter(df["biofilm_OD570"], df["metabolic_OD490"])
plt.xlabel("Biofilm biomass OD570")
plt.ylabel("Metabolic activity OD490")
plt.title("Biofilm Biomass vs Metabolic Activity")
plt.grid(True)
plt.show()

```





### Interpretation:

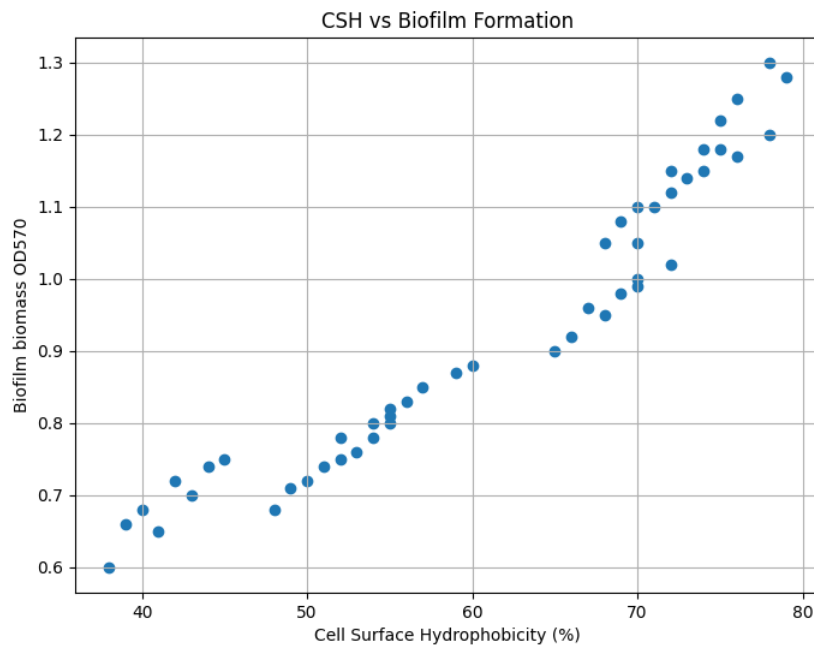
The scatter plot showed a positive relationship between biofilm biomass and metabolic activity. Isolates with higher biofilm OD570 values generally also had higher metabolic activity OD490 values.

---

### Figure 3. CSH vs biofilm

#### Code used:

```
plt.figure(figsize=(8,6))
plt.scatter(df["CSH_percent"], df["biofilm_OD570"])
plt.xlabel("Cell Surface Hydrophobicity (%)")
plt.ylabel("Biofilm biomass OD570")
plt.title("CSH vs Biofilm Formation")
plt.grid(True)
plt.show()
```



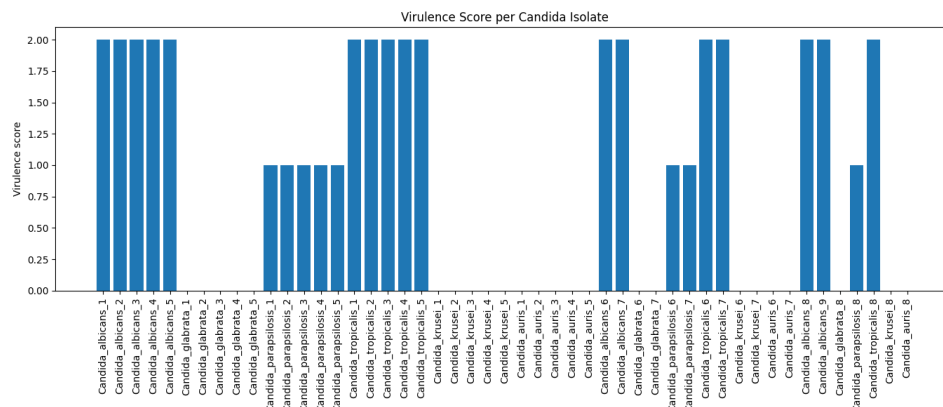
### Interpretation:

The graph showed that isolates with higher cell surface hydrophobicity generally had higher biofilm biomass.

## Figure 4. Virulence score per isolate

### Code used:

```
plt.figure(figsize=(14,6))
plt.bar(df["species"], df["virulence_score"])
plt.xticks(rotation=90)
plt.ylabel("Virulence score")
plt.title("Virulence Score per Candida Isolate")
plt.tight_layout()
plt.show()
```



### Interpretation:

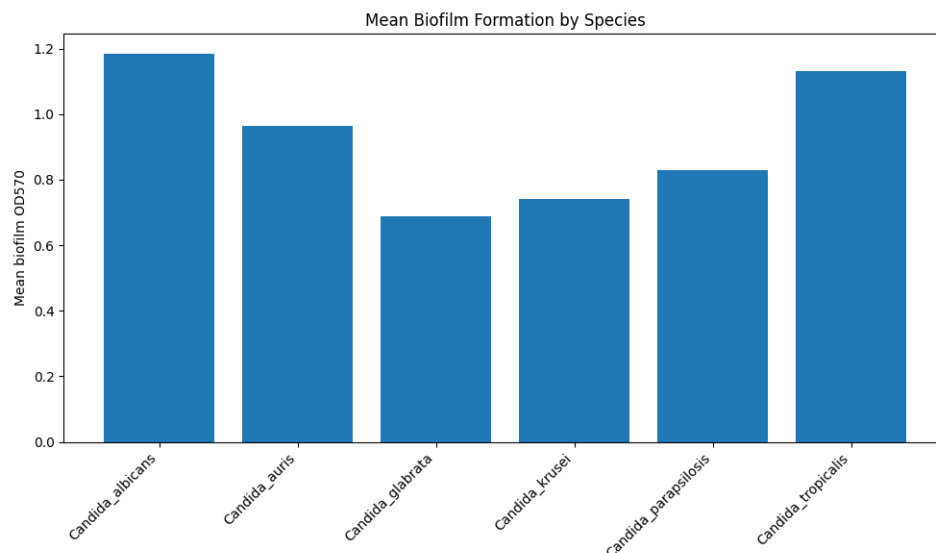
The virulence score varied between isolates. *Candida albicans* and *Candida tropicalis* showed the highest scores.

---

**Figure 5. Mean biofilm formation by species**

### Code used:

```
plt.figure(figsize=(10,6))
plt.bar(mean_by_species.index, mean_by_species["biofilm_OD570"])
plt.xticks(rotation=45)
plt.ylabel("Mean biofilm OD570")
plt.title("Mean Biofilm Formation by Species")
plt.tight_layout()
plt.show()
```



### Interpretation:

*Candida albicans* showed the highest mean biofilm formation.

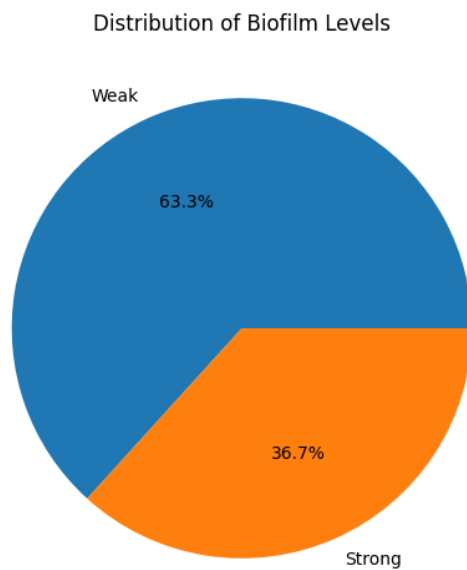
---

**Figure 6. Distribution of biofilm levels**

### Code used:

```
biofilm_counts = df["biofilm_level"].value_counts()

plt.figure(figsize=(6,6))
plt.pie(biofilm_counts, labels=biofilm_counts.index, autopct="%1.1f%%")
plt.title("Distribution of Biofilm Levels")
plt.show()
```



**Interpretation:**

63.3% weak and 36.7% strong producers.

---

## 6. General Interpretation

The interpretation of results aims to explain the biological significance of the observed patterns in *Candida* isolates.

Overall, the results showed that biofilm formation, metabolic activity, cell surface hydrophobicity, and virulence score varied between *Candida* species. *Candida albicans* and *Candida tropicalis* showed the highest mean biofilm values and virulence scores, suggesting that these species may have a stronger ability to form biofilms and carry selected virulence-associated genes.

The positive relationship between biofilm biomass and metabolic activity suggests that isolates forming more biomass also tend to contain more active cells. In addition, the relationship between CSH and biofilm formation suggests that cell surface hydrophobicity may play an important role in adhesion and biofilm development.

The phylogenetic analysis showed genetic relationships between selected *Candida* species based on the provided DNA sequences. The tree grouped species according to sequence similarity, with closer branches representing more similar sequences.

---

## 7. Conclusion

This Python project demonstrated how biological data can be analyzed using computational tools. The script allowed data loading, cleaning, calculation of new variables, graphical representation, sequence alignment, genetic distance calculation, phylogenetic tree construction, and export of results.

The analysis showed that *Candida albicans* and *Candida tropicalis* had higher biofilm formation and virulence scores compared with other species. The graphical results also suggested a positive association between biofilm formation, metabolic activity, and cell surface hydrophobicity. Finally, the phylogenetic tree provided a visual representation of genetic relationships between selected *Candida* species.

---